

## Activity 2: (Homework)

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Q1: Let  $\mathbf{H} = x^2 y \mathbf{a}_x + xy^2 \mathbf{a}_y$ .

Evaluate the line integral along the path  $x = y$  joining point  $P(1, 1, 0)$  to point  $Q(2, 2, 0)$ .

$$d\mathbf{r} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

So now the line integral becomes:

$$\int_C \mathbf{H} \cdot d\mathbf{r} = \int_C (x^2 y dx + xy^2 dy)$$

along  $x=y$ , we have  $dx=dy$ . Substituting  $y=x$   
and  $dy=dx$

$$\mathbf{H} \cdot d\mathbf{r} = (x^2 \cdot x dx) + (x \cdot x^2 dx) = x^3 dx + x^3 dx = 2x^3 dx$$

Now integrating from  $x=1$  to  $x=2$   
 $\uparrow \quad \quad \uparrow$   
 $(1, 1, 0) \quad (2, 2, 0)$

$$I = \int_1^2 2x^3 dx = \left[ \frac{2x^4}{4} \right]_1^2 = \frac{x^4}{2} \Big|_1^2$$

$$I = \frac{2^4}{2} - \frac{1}{2} = 8 - 0.5 = \boxed{7.5} \text{ Answer.}$$

Q2: Let  $\mathbf{A} = y\mathbf{a}_x + z\mathbf{a}_y + x\mathbf{a}_z$ . Find the flux of  $\mathbf{A}$  through surface  $y = 1$ ,  $0 < x < 1$ ,  $0 < z < 2$ .

Hint for Q2: You have to solve

$$\iint \mathbf{A} \cdot d\mathbf{S}$$

where  $d\mathbf{S} = dx dz \mathbf{a}_y$ .

We evaluate the surface integral:

$$\Phi = \iint_S \mathbf{A} \cdot d\mathbf{S}$$

$$\therefore d\mathbf{S} = dx dz \mathbf{a}_y$$

$$\text{So, } \mathbf{A} \cdot d\mathbf{S} = (y\mathbf{a}_x + z\mathbf{a}_y + x\mathbf{a}_z) \cdot (dx dz \mathbf{a}_y)$$

$$\mathbf{A} \cdot d\mathbf{S} = z dx dz$$

Now

$$\Phi = \int_0^2 \int_0^1 z dx dz$$

$$\phi = \int_0^1 \int_0^1 z \, dx \, dz$$

$$\phi = \int_0^2 x \Big|_0^1 z \cdot dz$$

$$\phi = \int_0^2 z \cdot dz = \frac{z^2}{2} \Big|_0^2 = \frac{4}{2} - \frac{0}{2} = \boxed{2}$$

flux of A through the surface is 2.